
EE1060: Differential Equations and Transform Techniques

Lecture 36: Homogeneous Systems of Differential Equations

Topics :

1. Homogeneous Systems of Differential Equations
 2. Wronskian and Fundamental set of Solutions
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Systems of Differential Equations, Existence & Uniqueness and Phase Plane Trajectories

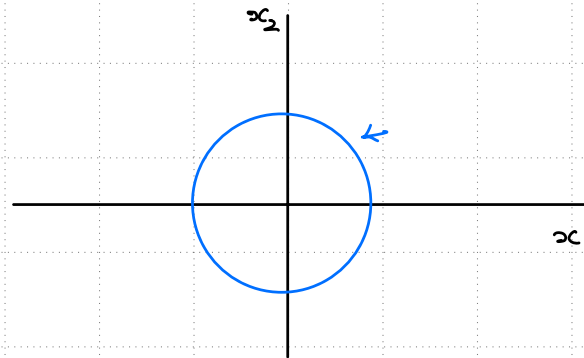
System of first-order ODE :-

$$\dot{\vec{x}} = A(t) \vec{x} + \underbrace{B(t) \vec{u}}_{\vec{f}} \rightarrow \textcircled{1}$$

- Recap:-
- a) $\vec{x}_s$ is a solution to $\textcircled{1}$ on I if $\vec{x}_s$ is differentiable on I & satisfies $\textcircled{1}$.
 $\nearrow \in \mathbb{R}$ (time)
 - b) Trajectory $\rightarrow$ has $\vec{x}_s$ evolves over it
 $\searrow \in \mathbb{R}^n$

Phase-plane.

Ex: if $\vec{x} \in \mathbb{R}^2$



Ex:- $\begin{bmatrix} \cos t \\ \sin t \end{bmatrix}$

if $\textcircled{1}$ corresponds to an ODE of order 'n' then

$$\vec{x} = \begin{bmatrix} x \\ x' \\ \vdots \\ x^{(n-1)} \end{bmatrix}$$

Existence & Uniqueness:-

for $\dot{\vec{x}} = A\vec{x} + \vec{f}$ with IC: $\vec{x}(t_0) = \vec{x}_0$

where $A(t)$ & $f(t)$ are continuous on I ,

then a unique sol exists.

Homogeneous Systems of Differential Equations and Wronskian

A system of ODE $\dot{\vec{x}} = A\vec{x} + B\vec{u}$ is homogeneous if

a) $\vec{u} \equiv \vec{0}$

b) $\vec{u} \in \text{Null space of } B$.

Homogeneous if: $\alpha\vec{x}$ is also a solution when $\vec{x}$ is a solution.

Let $\{\vec{x}_1, \dots, \vec{x}_n\}$ be sol of Homogeneous system of ODE $\dot{\vec{x}} = A\vec{x}$
 $\rightarrow$ LD set if $\sum_{i=1}^n \alpha_i \vec{x}_i = \vec{0}$ for some non-zero α_i
 $\rightarrow$ LI if $\sum_{i=1}^n \alpha_i \vec{x}_i = \vec{0}$ only when $\alpha_i = 0 \forall i$

for $\{x_1, \dots, x_n\}$ sol's of $L[y] = 0$, $W = \begin{vmatrix} x_1 & \dots & x_n \\ x_1' & \dots & x_n' \\ \vdots & & \vdots \\ x_1^{(n-1)} & \dots & x_n^{(n-1)} \end{vmatrix}$
 $\downarrow$ when converted into $\dot{\vec{x}} = A\vec{x}$
 $\vec{x}_1 \dots \vec{x}_n$

for gen $\dot{\vec{x}} = A\vec{x}$, we define $X = [\vec{x}_1 \dots \vec{x}_n]^T \xrightarrow{(+)} W(X)(t_0) = \det(X(t_0))$

Properties of Wronskian and Existence of Fundamental Set

- ① Wronskian $W \equiv 0$ over I or non-zero on I .
- ② $W(x) \equiv 0$ for a collection of LD set of solutions.
- ③ The set of sol's $\{\vec{x}_1, \dots, \vec{x}_n\}$ are LI if $W(x) \neq 0$ for some $t_0 \in I$.

fundamental set of sol's for $\dot{\vec{x}} = A\vec{x}$:- The set of LI sol's of $\dot{\vec{x}} = A\vec{x}$ is called the fundamental set.

if A is an $n \times n$ matrix
 $\hookrightarrow$ max. no. of elem in the fundamental set $\equiv n$.

A fundamental set of sol's always exists for $\dot{\vec{x}} = A\vec{x}$:-

Let $\vec{x}_i$ be the sol to IVP: $\dot{\vec{x}} = A\vec{x}$ with $\vec{x}(t_0) = \vec{e}_i \Rightarrow X = [\vec{x}_1 \dots \vec{x}_n]$

$W(x)(t_0) = \det[\vec{e}_1, \dots, \vec{e}_n] = 1 (\neq 0) \Rightarrow \vec{x}_1, \dots, \vec{x}_n$ are LI & hence they form a fundamental set of sol.

Solution to IVP - Linear Combination of Fundamental Set

The solution of $\dot{\vec{x}} = A\vec{x}$ with IC $\vec{x}(t_0) = \vec{x}_0$ can be expressed as a linear combination of the fundamental set i.e.,

$$\vec{\phi}(t) = \sum_{i=1}^2 \alpha_i \underbrace{\vec{x}_i}_{\text{elem of fundamental set.}}$$

The solution of $\dot{\vec{x}} = A\vec{x}$ with IC $\vec{x}(t_0) = \vec{x}_0$ can be expressed as a LC of the fundamental set

$$\vec{\phi}(t) = \sum_{i=1}^2 \alpha_i \vec{x}_i$$